MIL-B-83769/5C
4 April 1983
SUPERSEDING
MIL-B-83769/5B
15 September 1978

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, LEAD-ACID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery described herein shall consist of this document and the latest issue of the specification MIL-B-83769.

Military part number	-M83769/5-1 (see Figures 1 & 2) Not for Navy Use
	-M83769/5-2 (see Figures 3 & 4) For Navy Use
Nominal voltage	-24 volts
Maximum weight	-80 pounds (wet)
Capacity at 1-hour rate	-31 AH
Discharge rate for cell terminal post, intercell connectors, terminal connectors	-330 amps
Dimensions and configurations	-see Figures 1 through 4
Disconnect receptacle	-Figure 3 of MIL-B-83769

Notes:

1. Dimensions in inches, tolerance $\pm$0.063 unless otherwise specified.
2. Draft angles allowed in conformance with commercial die casting practices.

FSC 6140

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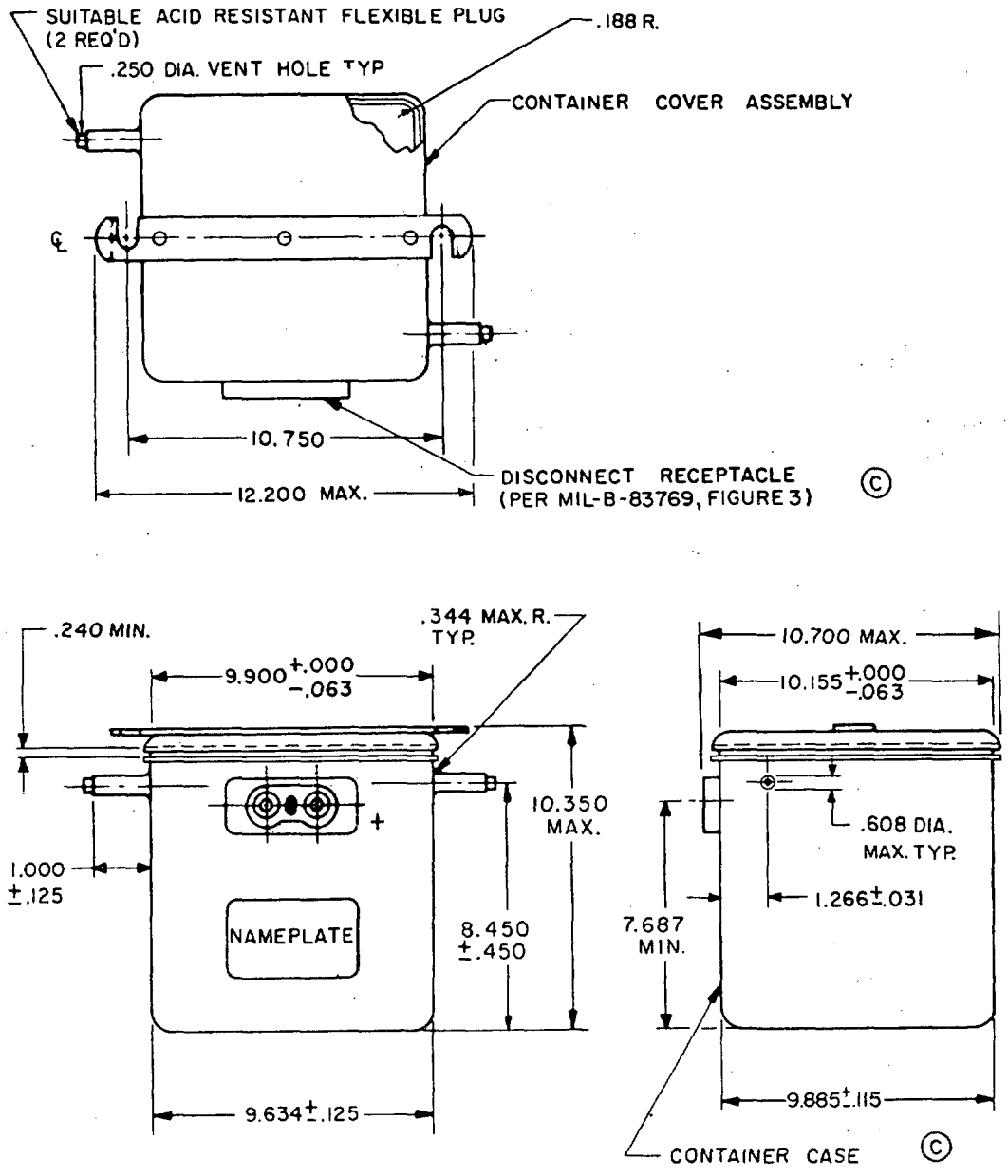


Figure 1. Battery container.

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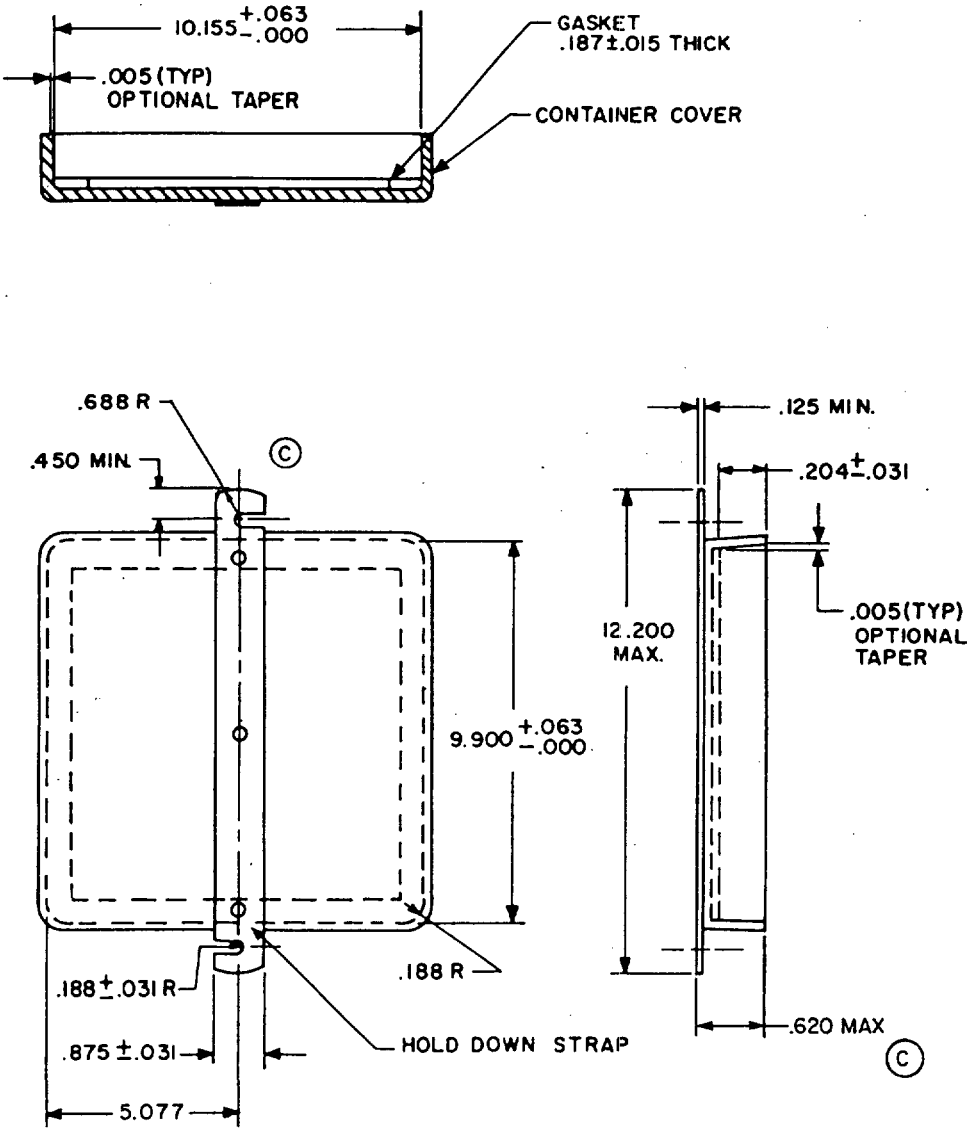


Figure 2. Container cover for Part No. M83769/5-1.

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M83769/5-2 shall conform to MIL-B-83769 except as modified by the following:

(C) 3.4.14 Interior and exterior finish. The exterior of metallic battery containers, covers, the interior and exterior of metallic disconnect receptacles and screws, except threaded areas, shall be thoroughly covered with an acid resistant coating which shall meet the requirements of 3.5.1 when subjected to the tests of 4.6.1.

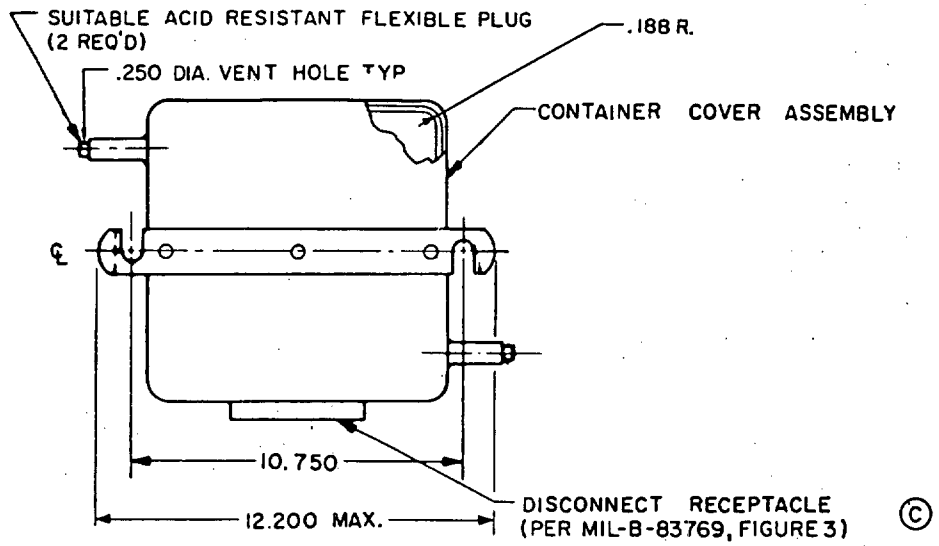
(C) 3.5.17 Life cycling. Batteries, when cycled in accordance with 4.6.18, shall provide no less than one hundred thirty (130) cycles of charge and discharge in an ambient temperature of $27^{\circ} \pm 5^{\circ}$C with the battery exposed on all four sides and the top (lid) off to allow free air circulation. During the life test, inability to meet the 1 hour rate discharge requirement for three (3) consecutive discharges shall constitute a failure. No reduction of capacity to less than 100% of that specified in Table I for cycles 126 thru 130 shall be permitted. During life cycling, batteries shall operate normally in an upright position, but shall be inverted for 2.5 minutes of discharge during cycle one hundred thirty (130). The battery shall not exhibit an excessive loss of electrolyte or a change in battery voltage in excess of 0.5 volt as a result of the inversion during cycle one hundred thirty (130).

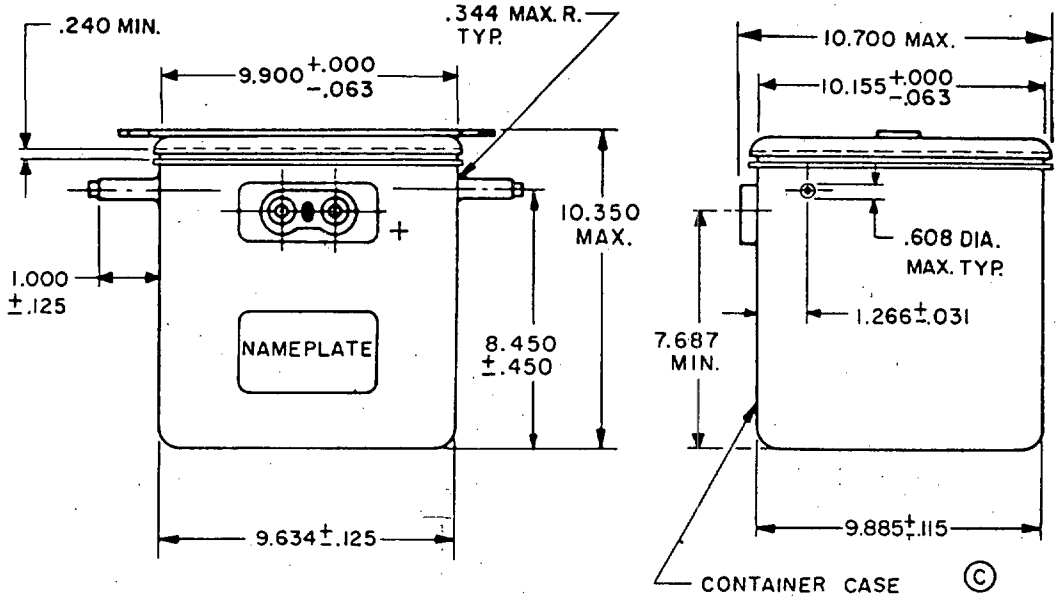
(C) 4.6.18.1 The charges preceding cycles 1 through 7 and 126 through 130 shall be in accordance with 4.5.1.3.1. The remaining charges during cycling shall be in accordance with 4.5.1.3.2.

4.6.18.2 Cycling. Cycling shall consist of a series of cycles of charge and discharge as specified below, all of which shall be counted as life cycles. Cycling shall be as follows:

Cycle Number	Test	Rate
1-3	1-hour at $27^{\circ} \pm 5^{\circ}$C	1.0 C
4	5-minute at $27^{\circ} \pm 5^{\circ}$C	6.0 C
5	5-hour at $27^{\circ} \pm 5^{\circ}$C	0.2 C
6	5-minute at $-18^{\circ} \pm 2^{\circ}$C	6.0 C
7-125	1-hour at $27^{\circ} \pm 5^{\circ}$C	1.0 C
126	5-minute at $27^{\circ} \pm 5^{\circ}$C	6.0 C
127	2-hour at $27^{\circ} \pm 5^{\circ}$C	0.5 C
128	5-minute at $27^{\circ} \pm 5^{\circ}$C	6.0 C
129	2-hour at $27^{\circ} \pm 5^{\circ}$C	0.5 C
130	5-minute at $27^{\circ} \pm 5^{\circ}$C	6.0 C

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MATERIAL: NON-METALLIC, ALUMINUM, STEEL OR ALUMINUM ALLOY
DIE CASTING PER QQ-A-591

PART NO.
M83769/5-2
FOR NAVY USE

Figure 3. Battery container.

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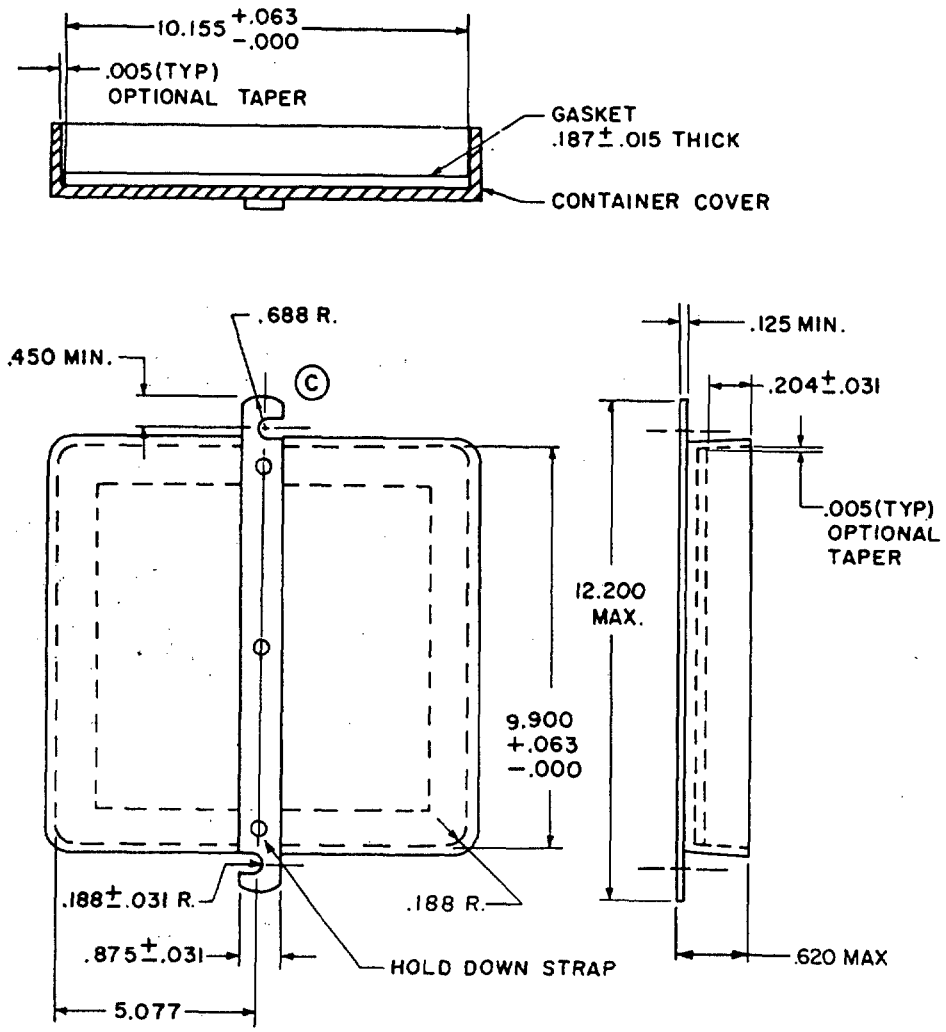


Figure 4. Container cover for Part No. M83769/5-2.

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During cycles 7 thru 125, the batteries shall not require water additions more than once in four (4) consecutive cycles. During life cycle one hundred thirty (130), when the battery is subjected to the five (5) minute discharge, the battery shall be inverted during the last 2.5 minutes of discharge.

International standardization. Certain provisions of this specification are subject to International Standardization Agreements (ASCC AIR STD 12/15 and STANAG 3514AE). When revision or cancellation of this standard is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including department standardization offices, if required.

Custodians:
Air Force - 99
Army - ER
Navy - AS

Preparing activity:
Navy - AS
(Project No. 6140-0572-5)

Review activities:
Air Force - 11, 80
Army - AV
DLA - GS

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